

1. **Scenario:** A user is required to enter a valid number in a form, but users sometimes input invalid data.

Write logic to repeatedly prompt the user until they enter a valid integer.

- Get a user valid input number.
- Try converting to integer.
- Write an if else condition to get a prompt.
- If valid - print valid number.
- Else - print invalid number and repeat.

2. **Scenario:** A data analysis tool processes a list of numbers and needs to identify the most frequently occurring value.

Write logic to find the most frequently occurring number in a given list.

- Get a list of numbers.
- Count occurrences using for loop.
- Using if condition to find number with highest count.

3. **Scenario:** A text-processing application needs to compare words and check if they are anagrams (contain the same letters in a different order).

Write logic to determine whether two given strings are anagrams.

- Get an input of the two words.
- Sort both strings, Compare results.
- And using If else condition to print as given strings are anagrams.
- Or given strings are not an anagram.

4. **Scenario:** A speech analysis program needs to count the number of vowel sounds in a given input.

Write logic to count the number of vowels in a given string.

- Get an input string word.
- Convert the string into lowercase using lower().
- Define vowels. Using a loop, check each character in the string.
- If the character is a vowel, increase the count, Get an output number of vowels.

5. **Scenario:** A text-editing software includes a feature to reverse the order of words in a sentence for stylistic effects.

Write logic to reverse the order of words in a sentence while keeping the words themselves intact.

- Get a sentence as string input.
- Split the sentence into words.
- Using reverse and print the reverse sentence.

6. **Scenario:** A missing number is detected in a sequence of values stored in a database.

Write logic to find the missing number in a list containing $n-1$ numbers from 1 to n .

- Get a list containing $n-1$ numbers.
- Find the value of n (total numbers expected).
- Calculate the sum of numbers from 1 to n .
- Calculate the sum of elements in the given list.
- Subtract list sum from total sum.
- Print the missing number.

7. **Scenario:** An ATM machine processes withdrawal requests and needs to ensure that users cannot withdraw more than their account balance.

Write logic to allow a withdrawal only if the balance is sufficient.

- Get account details and withdrawal amount.
- Compare withdrawal amount with balance using if-else condition.
- If withdrawal amount is less than or equal to balance, print balance sufficient.
- Else, print insufficient balance.

8. **Scenario:** A system needs to verify whether a given dataset contains duplicate entries.

Write logic to check whether a given list contains duplicate values.

- Get a given list.
- Convert the list into a set.
- Compare the length of the list and the set.
- If both lengths are different, print duplicate values found.
- Else, print no duplicate values.

9. **Scenario:** A digital calculator includes a feature to sum the digits of a number for verification purposes.

Write logic to calculate the sum of all digits in a given integer.

- Get an integer number as input.
- Initialize sum value as 0.
- Using a loop, extract each digit of the number.
- Add each digit to the sum.
- Remove the last digit after each iteration.
- Print the final sum of digits.

10. **Scenario:** A language-learning app wants to verify whether a given sentence is a pangram (contains every letter of the alphabet at least once).

Write logic to check if a given sentence is a pangram.

- Get a sentence as input.
- Convert the sentence into lowercase.
- Remove spaces and special characters if needed.
- Check whether all alphabets from a to z are present.
- If all letters are present, print sentence is a pangram.
- Else, print sentence is not a pangram.